
Appendix

Topology of Cell Complexes

Here we collect a number of basic topological facts about CW complexes for convenient reference. A few related facts about manifolds are also proved.

Let us first recall from Chapter 0 that a CW complex is a space X constructed in the following way:

- (1) Start with a discrete set X^0 , the 0-cells of X .
- (2) Inductively, form the n -skeleton X^n from X^{n-1} by attaching n -cells e_α^n via maps $\varphi_\alpha: S^{n-1} \rightarrow X^{n-1}$. This means that X^n is the quotient space of $X^{n-1} \amalg_\alpha D_\alpha^n$ under the identifications $x \sim \varphi_\alpha(x)$ for $x \in \partial D_\alpha^n$. The cell e_α^n is the homeomorphic image of $D_\alpha^n - \partial D_\alpha^n$ under the quotient map.
- (3) $X = \bigcup_n X^n$ with the weak topology: A set $A \subset X$ is open (or closed) iff $A \cap X^n$ is open (or closed) in X^n for each n .

Note that condition (3) is superfluous when X is finite-dimensional, so that $X = X^n$ for some n . For if A is open in $X = X^n$, the definition of the quotient topology on X^n implies that $A \cap X^{n-1}$ is open in X^{n-1} , and then by the same reasoning $A \cap X^{n-2}$ is open in X^{n-2} , and similarly for all the skeleta X^{n-i} .

Each cell e_α^n has its **characteristic map** Φ_α , which is by definition the composition $D_\alpha^n \hookrightarrow X^{n-1} \amalg_\alpha D_\alpha^n \rightarrow X^n \hookrightarrow X$. This is continuous since it is a composition of continuous maps, the inclusion $X^n \hookrightarrow X$ being continuous by (3). The restriction of Φ_α to the interior of D_α^n is a homeomorphism onto e_α^n .

An alternative way to describe the topology on X is to say that a set $A \subset X$ is open (or closed) iff $\Phi_\alpha^{-1}(A)$ is open (or closed) in D_α^n for each characteristic map Φ_α . In one direction this follows from continuity of the Φ_α 's, and in the other direction, suppose $\Phi_\alpha^{-1}(A)$ is open in D_α^n for each Φ_α , and suppose by induction on n that $A \cap X^{n-1}$ is open in X^{n-1} . Then since $\Phi_\alpha^{-1}(A)$ is open in D_α^n for all α , $A \cap X^n$ is open in X^n by the definition of the quotient topology on X^n . Hence by (3), A is open in X .

A consequence of this characterization of the topology on X is that X is a quotient space of $\amalg_{n,\alpha} D_\alpha^n$.